The Future of ClimateWins: Predicting Weather Conditions

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Objective

Objective: Predict future weather in mainland Europe by using historical weather data with machine learning methods.

Why? Increasing temperature and weather events has become more common in recent times. ClimateWins hopes to be able to predict these changes to help predict future weather patterns to improve forecasting and preparedness.

Method: Machine learning will allow ClimateWins to leverage pattern recognition often found in weather data to help predict future weather patterns.

Why Machine Learning?

Unsupervised Learning

Principal Component Analysis (PCA)
- Reduces high-dimensional climate data into key components
- Highlights dominant weather patterns, improves efficiency and interpretability

K-Means & Hierarchical Clustering
- Identify natural groupings in weather behavior without labels

Deep Learning

Convolutional Neural Networks (CNNs)
- Uses various hidden layers to learn patterns in the data for prediction

Detecting evolving weather systems across Europe

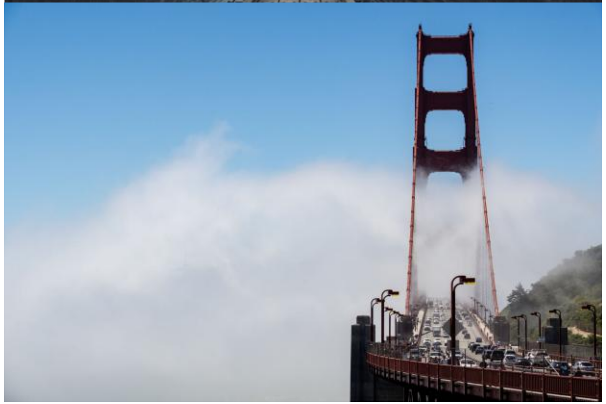
Recurrent Neural Networks (RNNs) & Long Short-Term Memory (LSTM)
- Model how past conditions influence future outcomes
- Essential for forecasting and trend detection

Random Forests
- Provides feature importance, improving model transparency through a collection of decision trees

Generative Adversarial Networks (GANs)
- Use two competing neural networks to generate artificial data to help train thee model.
- Use photos like the ones on the right to predict current and future weather conditions

Optimization

Bayesian Optimization
- Efficiently tunes hyperparameters across all models



Thought Experiments

Real-Time Analysis

What is current weather like? Identify current weather patterns using a mixture of historical data, current weather conditions and real-time photos.

Weather Alerts

What does weather in the near future look like and do people need to be alerted of dangerous weather patterns? Alert of unusual patterns that are happening now and the future allowing citizens and first responders to properly prepare.

Future Safe Living

Where are the safest regions for citizens to live in the next 25-50 years? Use machine learning to determine the safest places for people to live in the next 25-50 years.

Identifying Current Weather Patterns

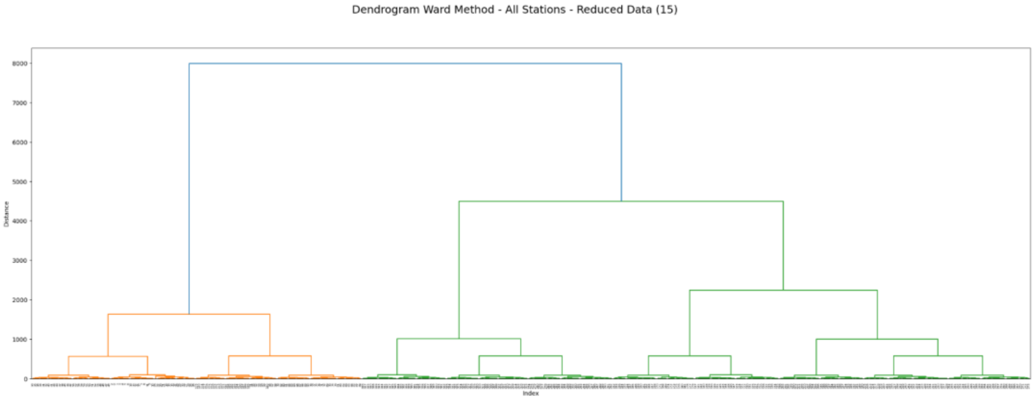
Objective: Use historical and real-time data to recognize emerging weather patterns across Europe and more accurate daily forecasts.

Data Needed

- Historical daily weather data from 1960-2022
- Real-time weather station feeds

Machine Learning Methods

- **Principal Component Analysis:** Reduce dimensionality and highlight dominant climate trends
- **K – Means:** Group days or regions with similar weather behavior
- **Hierarchical Clustering:** Reveal relationships between weather patterns through dendograms



Example above has two clusters for pleasant and unpleasant weather. This could further be trained to include more classifications such as sunny, rainy, cloudy, etc.

Output

- Classification of current weather states
- Detection of transitions toward extreme conditions
- Discovery of recurring seasonal and regional weather patterns

Predicting and Alerting on Unusual Weather Events

Objective: Detect anomalous weather behavior and provide early warnings.

Data Used/Needed

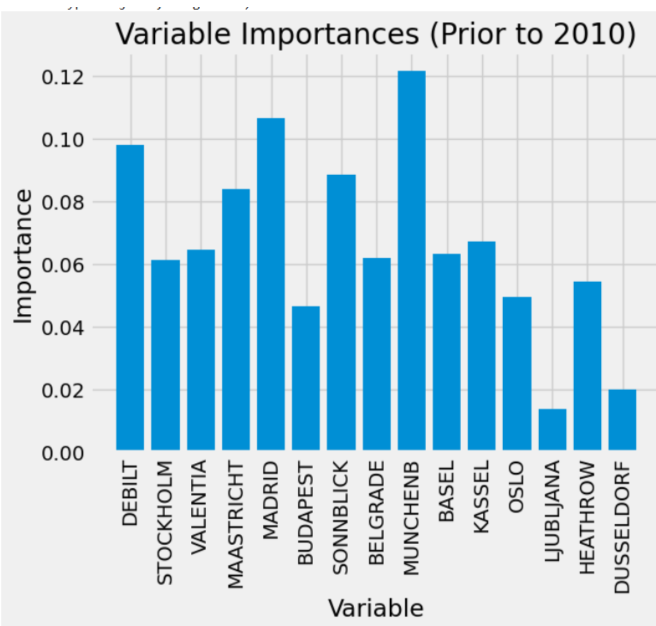
- Historical climate data which contains precipitation levels, average temperatures and other basic weather information etc.
- Historical extreme weather records
- Real-time temperature, wind, and precipitation data

Machine Learning Methods

- **Random Forests:** Classify and predict extreme weather events using historical patterns
- **Recurrent Neural Networks:** Capture short-term temporal dependencies in weather sequences
- **Long Short-Term Memory LSTM:** Model long-term dependencies and delayed effects in climate trends
- **Bayesian Optimization**: Tune model hyperparameters to improve predictive accuracy

Output

- Early warnings for extreme heat, storms, or unusual cold events
- Risk scores and confidence intervals for decision-makers



Identifying Safest Regions for Long-Term Living (25–50 Years)

Objective: Predict which regions are most likely to remain climatically stable and livable.

Data Used

- Long-term temperature and precipitation trends
- Frequency of extreme events

Machine Learning Methods

- **Random Forests** for feature importance
- **Generative Adversial Networks (GANs)** Generate future weather climate scenarios
- **PCA + Clustering** Reduce complexity and group regions with similar climate trajectories

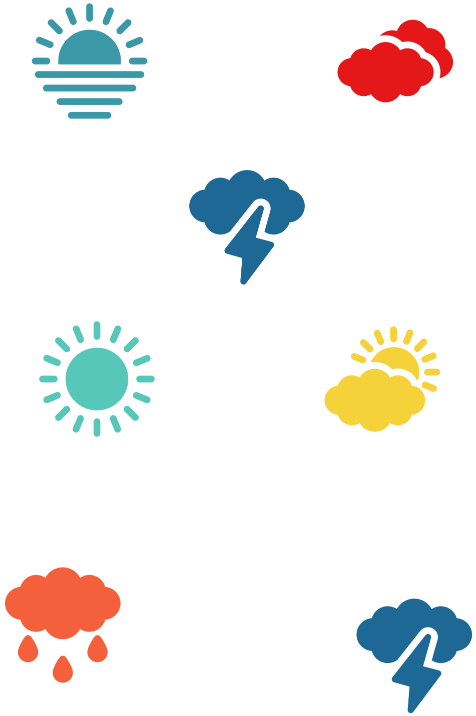
Output

Regional climate scores

Support for urban planning and policy decisions

Correct Prediction - class: Rain - predicted: Rain[1.2001851e-08 1.0000000e+00 1.2689975e-09 8.4344497e-11]

Conclusion



Machine learning enables us to take complex datasets and understand climate behavior at a deeper level.

- A combination of models typically provides a reliable model – Ensemble approach

Thank you!

QUESTIONS?

GITHUB: ISAVANNAHR/CF_CLIMATEWINS

VIDEO: HTTPS://YOUTU.BE/0ZEK45DZVHS